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Augmenting, Not Replacing: The Role of LLMs in Human-Centric Formal RE

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***Abstract*—Formal methods for requirements engineering have existed for decades; yet, these techniques are rarely used if not required by certification because they are challenging for non-experts (e.g., novices and non-technical stakeholders in multi-disciplinary teams) to interpret and apply. To enable non-experts to participate in collaborative software teams, we envision using artificial intelligence (AI) to assist in interpreting formal notations. Our research project investigates how and to what extent generative AI with large language models (LLMs) can be used to assist non-experts in interpreting formal requirements. In this paper, we conduct an exploratory investigation of both generating translations and interpreting linear temporal logic (LTL) formulae. Specifically, we explore prompting LLMs with sufficient information for the task of generating LTL formula explanations. With our initial prompt, we complete a classroom study where students learn LTL and then interpret a series of LTL formulae with and without the LLM-generated descriptions. We then improve our approach based on insights from the classroom study, and evaluate the overall quality of our updated prompt and the explanations it generates.**

I. Introduction

Formal methods (FMs) and formal notations have been long established as powerful tools in requirements engineering (RE). The primary advantage of FMs is their ability to provide a mathematically precise specification, reducing ambiguity and enabling early verification of requirements. Techniques such as model checking, theorem proving, and formal refinement allow for automated analysis, reducing the risk of defects in the early stages of system development [1]. Moreover, formal specifications can facilitate reasoning about correctness properties, ensuring that critical system requirements (i.e., safety and security) are rigorously validated before implementation [2].

Despite their benefits, the mathematical nature of FMs can make them difficult for non-specialists to understand [3], [4], [5]. In practice, this leads to a gap between formal specifications and real-world applications, particularly when communicating with non-technical stakeholders. To address this, hybrid approaches that combine FMs with semi-formal techniques (e.g., controlled natural languages and graphical representations), have been explored to improve usability while retaining precision [6]. However, even in hybrid approaches the resulting specifications can remain opaque to the stakeholders (i.e., the people who need to apply them), creating a gap between the formal notations and their domain understanding.

Motivating Example. Consider the operations of a microwave: *start*, *cook*, and *stop*. One can write a Linear Temporal Logic (LTL) formula (see Sect. II for an overview) to describe the function of a microwave as in Def. 1, where once *start* is activated, the microwave should *cook* until the user presses *stop*. Yet, a non-expert stakeholder may give the same meaning to Def. 2 as Def. 1, not realizing the critical role of the parentheses. In Def. 2 the global operator G only applies to *start* and there is no connection with the consequent of the implication. Further, in Def. 2, the consequent of the until operator U (i.e., *stop*) depends on a future state of the antecedent (i.e., X *cook*). With explanations, we can help individuals better interpret the correctness of a formula.

$$G\ (start \implies X\ (cook\ U\ stop)) \tag{1}$$

$$G\ start \implies X\ cook\ U\ stop \tag{2}$$

State of the Art. Recent advances in generative artificial intelligence (AI), particularly large language models (LLMs), offer promising opportunities to bridge this divide between experts and stakeholders [7]. These models have demonstrated remarkable capabilities in understanding context, generating natural language explanations, and even reasoning about formal systems [8]. However, their potential as assistive tools for interpreting formal requirements remains largely unexplored.

Studies have shown that LLMs can successfully transform unstructured natural language into formal specifications [9], [10]. Additionally, LLMs have been assessed for their ability to verify logical statements and reason about formal languages, with results indicating that while they approach human-level comprehension, LLMs struggle with accurately generating logical forms [11]. Another use of LLMs is to advance theorem proving to maximize the number of automatically generated proofs. For example, a recent tool called Rango integrates LLMs, with retrieval-augmented proving, to enhance proof synthesis within the Coq proof assistant [12]. Rango works by dynamically retrieving relevant premises and similar proofs during the proof process, tailoring the context for the LLM and significantly improving the proof automation process. However, the approach does not provide any explanations for users, and thus, forces users to naively trust the result.

Ferrari and Spoletini created a roadmap for LLMs and formal requirements engineering (RE) [8] that emphasizes the

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dual role of LLMs in both automating key RE tasks (e.g., elicitation, analysis, and validation), and improving the interpretability of formal specifications. By using their advanced natural language processing capabilities, LLMs can mitigate common obstacles in formal RE, including ambiguity and inconsistency, while making FMs more accessible to practitioners and stakeholders. Furthermore, the roadmap highlights the potential of LLMs to bridge the gap between formal and natural language representations, thereby enhancing the usability of FMs in industry and research.

Problem. Despite these advancements, LLMs exhibit notable limitations. LLMs may produce plausible-sounding but incorrect interpretations, struggle with complex nested expressions, and fail to distinguish subtle semantic differences in formal notations, as the choice of prompt format, training examples, and even the order of the training examples can cause accuracy to vary [13]. These challenges highlight the importance of systematic evaluation and careful application when leveraging LLMs as assistive tools in FMs. Some research suggests that combining LLMs with traditional FMs could enhance their reliability and effectiveness in practical applications [14], [15].

Contributions. In this paper, we investigate how and to what extent generative AI with LLMs can be used to assist non-experts (i.e., stakeholders and novices) in interpreting formal requirements. In order to give context to our work and user studies, we use Linear Temporal Logic (LTL) [16] as a language to express formal requirements. LTL is widely used in formal RE and is highly applicable in software verification, hardware design, safety-critical systems, and robotics, ensuring that systems operate correctly over time. Despite this, many misconceptions exist about the interpretation of LTL and its English meaning [5].

Specifically, we explore prompting techniques to automatically generate explanations of LTL formulae. We curate a *gold standard* dataset of 50 LTL formulae and summaries. Using our initial prompt, we conduct a classroom study on the effectiveness of LLM-generated explanations in enhancing students' comprehension of LTL formulae. Building on the lessons learned from the classroom study, we iteratively improve our LLM prompt, focusing on clarifying students' confusion. Finally, we investigate the quality of the results of our updated prompt by linguistically comparing it with the original prompt, our classroom study, and our gold standard. This research addresses the longstanding challenge of making formal requirements more accessible without sacrificing their precision, with implications for education, industrial practice, and tool development in requirements engineering.

Organization. In the remainder of the paper, we review LTL and LLM prompting techniques in Sect. II. Sect. III introduces our initial prompt, the 50F dataset, and gold standard. Sect. IV evaluates the initial prompt through a classroom study. Sect. V describes our improved prompt. Sect. VI explores the quality of both prompts. Sect. VII discusses limitations and implications of this work. Sect. VIII positions our contributions within the literature. We conclude in Sect. IX.

II. Background

Linear Temporal Logic (LTL). LTL is a modal temporal logic that extends propositional logic with operators expressing properties over paths in time [16]. Introduced to reason about concurrent programs, LTL has become fundamental in formal verification, particularly for specifying behavioral properties of reactive and concurrent systems [2].

LTL formulae are built upon atomic propositions using logical operators (i.e., and &, or |, implication $\rightarrow$, not !) and temporal operators. We use four LTL temporal operators in our work:

Xa Next: The property a holds in the next state.
Fa Eventually: The property a will hold at some future state.
Ga Globally: The property a holds at every state.
aUb Until: The property a holds until b becomes true.

The semantics of LTL are defined over infinite sequences of states (traces), where each state represents a set of atomic propositions that hold true at that point. For example, the formula $G(\textit{request} \rightarrow F\ \textit{response})$ specifies that globally (at every state), if a *request* occurs, then eventually a *response* will follow—capturing behavioral properties in a reactive system.

Prompting Strategies. The interaction with LLMs is predominantly facilitated through *prompting*, where users provide specific information to the LLM to guide the model toward generating desired outputs. Several prompting strategies have emerged in the literature [17], [18], [19], [20]:

Zero-shot prompting: The model is asked to perform a task without examples.
One-shot prompting: The model is provided with a single example before performing the task.
Few-shot prompting: Multiple examples are provided to establish a pattern.
Chain-of-thought (CoT) prompting: The model is guided to break down complex reasoning into intermediate steps.
Tree-of-thought prompting: Extends chain-of-thought by exploring multiple reasoning branches.
Role prompting: The model is instructed to assume a specific role or persona; in this case, different LLM instances can assume distinct personas to engage in collaborative tasks.

For tasks involving formal notations (e.g., LTL), structured prompts have proven effective, as they use the model's ability to recognize patterns and follow defined rules [21], [18]. We use one-shot prompting, a guided decomposition strategy for eliciting reasoning, and role prompting strategies. Our approach differs from CoT prompting because we define an explicit structure for the reasoning steps within the prompt itself, guiding the model to follow our specified reasoning.

III. Preliminaries: Data and Prompt

LTL Notation Standardization. There are multiple syntactic variations to represent LTL across the academic literature [22]. To prioritize notational consistency across the multiple stages of our project, we adopt a uniform alphabetic format (i.e., F, G, U, and X) for temporal operators and character format (i.e., &, |, !, =>) for logical operators, rather than alternative

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symbolic representations (e.g., $\Diamond$, $\Box$, $\circ$, $\wedge$, $\vee$, and $\neg$). This standardization was intended to make the notation more accessible and easier to parse by both LLMs and human readers. In our classroom study (see Sect. IV), we needed students to be able to write LTL formulae in plain text. In subsequent analysis, we needed the formulae to appear naturally across multiple tools and file formats (e.g., .csv). A downside to this approach is that implication => required two characters. For readability, we use $\rightarrow$ for implication in what follows.

50F Dataset and Gold Standard. To systematically evaluate the effectiveness of our prompts, we created a testing corpus of 50 LTL formulae, which we call the *50F dataset* in this paper. We selected 25 formulae from the dataset by Cherukuri et al. [23] and wrote an additional 25 formulae based on educational texts that teach LTL (e.g., [24]). We curated a diverse set of LTL formulae representing a spectrum of complexity levels. The test set included basic logical operators, simple temporal operators, compound formulae, and complex nested expressions. This range in aspects of LTL syntax allows for a comprehensive evaluation of LLM capabilities in this domain.

Two authors collaborated to create natural language rephrasing of each formula in the 50F dataset. Each one-sentence *formula summary* acted as our gold standard of a human readable explanation of the LTL formula, see Sect. VII for a discussion of the limitations of our gold standard. Our 50F dataset and gold standards are available as part of our online open science supplement[1].

Prompt Structure. Our initial LTL prompt was intentionally designed as a preliminary approach, as per an iterative design paradigm [25], with the expectation that empirical insights from the classroom study (see Sect. IV) would expose areas of improvement. We developed a systematic prompting methodology to guide LLMs through a structured reasoning process when translating and explaining LTL formulae. The prompt template, listed in Fig. 1, was constructed to elicit explanations that break down LTL formulae into comprehensible parts.

This prompt structure integrates several methodological considerations (see Fig. 1). First, it implements the *one-shot* prompting paradigm (see Sect. II) as it is initiated with a single clear example translation to establish the expected output format, where the example is excluded from any test dataset, reflecting the nature of in-context learning which operates without parameter updates [26]. Simultaneously, a structured step-by-step reasoning approach is used by providing a predefined template within the prompt, which guides the model to break down the formula into components before constructing the meaning. This reflects established teaching methodologies for formal notations. Additionally, this prompt emphasizes interest in the reasoning process rather than the mere correctness of the translation, encouraging a more explanatory reasoning over black-box answers. Finally, a clear set of tasks was defined for the LLM, including using a bottom-up approach to construct the meaning of a formula from each of its constituent parts, mirroring how humans break down complex formulae.

[1]See *https://doi.org/10.35482/csc.004.2025* for supplement.

This is an example of an LTL translation to structured NL: X !e X f means "in the next moment of the system execution, e should not hold and, in the next moment of the system execution, f has to hold".
Please show me how you would reason to get a translation of what this formula means: <INSERT LTL FORMULA>
I am much more interested in understanding how to read it than reaching the correct answer. Do it in a concise way.
Follow this structure:
1. Write down original LTL formula
2. Break the subformula into a parsing tree (breakdown)
3. Rebuild meaning from the bottom of the tree up
4. Refine the natural language phrasing for clarity

Fig. 1. Original LLM LTL Prompt Template (LLM_Org)

LLM Selection. While the selection of a LLM impacts the performance of any generation task, the goal of this research is not to compare multiple LLMs. Thus, we informally tested our initial prompt (see Fig. 1) with three versions of OpenAI's LLMs: GPT-4o, their flagship model with 1.8 trillion parameters and advanced reasoning capabilities; GPT-4o-mini, a more efficient alternative with about 400 billion parameters; and GPT-o3-mini, which has around 175 billion parameters [27]. All models were presented with identical prompts and formula sets to ensure comparative validity. Among the models tested, GPT-4o demonstrated the best results in accurately interpreting and explaining LTL formulae. While all models followed our structured prompting approach, GPT-4o produced more concise and semantically correct explanations, particularly when handling complex nested temporal operators, which is helpful for non-expert users. GPT-4o was also more consistent in creating parse trees. For these reasons, we selected GPT-4o (i.e., *gpt-4o-2024-08-06*) for use in this paper. We used the default values for all parameters except temperature, which we lowered to 0.2 to achieve more deterministic results. Future work will compare LLMs (see Sect. IX for details).

Formula Explanations vs. Summaries. Within our prompts, we ask the LLM to follow a specific structure (see Fig. 1). In this paper, we define the formula *explanation* as the entire result produced by the LLM. Conversely, we define the formula *summary* as a natural language refinement of the formula (i.e., Step 4 in the prompt in Fig. 1), which is the last excerpt of the LLM's explanation. For example, Fig. 2 shows a lightly formatted LLM explanation. In this case, the formula explanation is the entire text of Fig. 2, whereas the summary is only the text immediately after "Refined phrasing:". Our gold standard described above contains only summaries for each formula. Similarly, in the classroom study (see Sect. IV), students were asked to write formula summaries via the prompt, "Please explain the meaning of the formula in your own words (1 sentence)".

Correctness of Initial Prompt. Using GPT-4o, we generated explanations and formula summaries for each of the formulae in our 50F dataset. We randomized the formulae and one author (an LTL expert) evaluated each summary for correctness.

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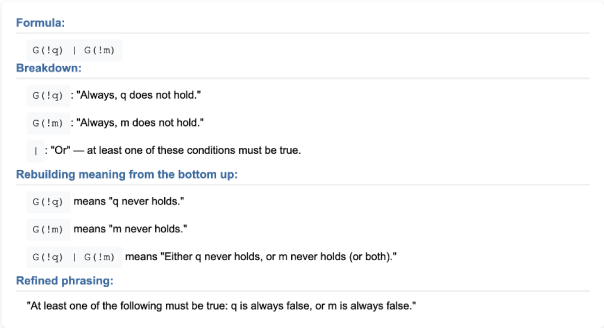

Fig. 2. LLM-based Explanation for $G(!q) \mid G(!m)$ using LLM_Org

94% (47/50) of the summaries were adjudicated as correct. We discuss additional analysis of our initial prompt in Sect. VI.

IV. Classroom Evaluation

In this section, we describe our study to explore to what extent the LLM explanations assisted the students in interpreting LTL formulae and whether students found them helpful.
In this investigation, we answered three research questions:

RQ1 Do LLM explanations affect students' understanding of LTL formulae?
RQ2 Do students find the explanations helpful?
RQ3 How do students perceive the LTL intervention?

A. Methodology

We conducted our investigation in February 2025 at Smith College, and our protocol was reviewed by our Institutional Review Board. Supplemental materials and study data are available online[1].

Study Rationale and Experimental Design. We investigated how LLM-based descriptions can assist non-experts in interpreting formal notations through a classroom study. Our evaluation was designed as an educational intervention with a single sequence. Each student participated in a short lecture (Period 0, see Table I), and then completed the study intervention. The intervention consisted of three periods. In Period 1, students received instructions on how to complete the survey instrument and reviewed a cheatsheet of logical and temporal operators. Students were given both physical and electronic copies of the cheatsheet for ease of reference in the remaining periods. In Period 2, students considered a series of eight LTL formulae, which are listed in Table I. In half of the cases (see LLM Explanation column in Table I), students were given the LLM explanation generated using the prompt we discuss in Sect. III. For example, Fig. 2 shows the explanation for the Formula F (in Table I) as it appeared in our study instrument. For each formula and explanation, students were asked 4-5 questions. In Period 3, students reflected on learning LTL and the explanations.

To create our questionnaire, we began with the dataset from Cherukuri et al. [23] and reviewed the work of van Lamsweerde [24]. From these sources, we curated a set of thirteen formulae that represent common temporal patterns, covering a range of logical and temporal operators. We discussed and selected eight of the original thirteen formulae, looking for pairs at a similar level of difficulty (see Table I for the final list). Given the threat of carryover and a learning effect, we varied the order of which formula received the explanation, as we increased the level of difficulty with each pair (i.e., A/B, C/D, E/F, and G/H). We then created a list of types of questions (e.g., applying a sequence to a formula), and varied the question types throughout each part. This initial educational intervention[1] (i.e., study instrument) is not complete in that it does not ask each type of question for each formula, but we made this design choice to ensure that our intervention with the short lecture would fit into a single 75-minute class block. Additionally, we wanted the students to not take the LLM explanations as correct, so for each explanation, we asked the question, "Do you think the explanation of the formula above is correct (Yes/No)? Why or Why not?", which we call the *LLM-Belief* question in the remainder of this paper.

TABLE I
Study Protocol & List of Formulae

Period	Details		
0	Introduce LTL		
1	Review Cheatsheet and Collect PII		
2	Answer Questions about Formulae		
	ID	Formula	LLM Explanation
	A	$G(!a)$	No
	B	$F(!e)$	Yes
	C	$G(!p \rightarrow X s)$	Yes
	D	$G(c \rightarrow X\ G\ r)$	No
	E	$F(!p\ \&\ !u)$	No
	F	$G(!q) \mid G(!m)$	Yes
	G	$G(!a \rightarrow F\ z)$	Yes
	H	$G(a \rightarrow F(v\ U\ w))$	No
3	Reflection		

Study Context and Participant Information. The evaluation was conducted during a 75-minute class session as part of a required second-year undergraduate computer science (CS) theory course. The previous and next course topics were pushdown automaton and Turing machines, respectively. The course instructor (see Acknowledgments) presented the introductory materials (i.e., the lecture in Period 0) similar to other classes, which required 25 minutes. A student researcher, acting as an in-class teaching assistant, was on hand to help answer and document student questions.

After the lecture, students were given access to a Qualtrics survey for the intervention in Periods 1–3. We originally envisioned the intervention instrument as a paper handout, but the course instructor recommended using Qualtrics because he had used it frequently for in-class activities—students were quite familiar with it. Students were encouraged to use other resources as part of the session and were able to ask questions of the instructor and the assistant. Students could leave at any

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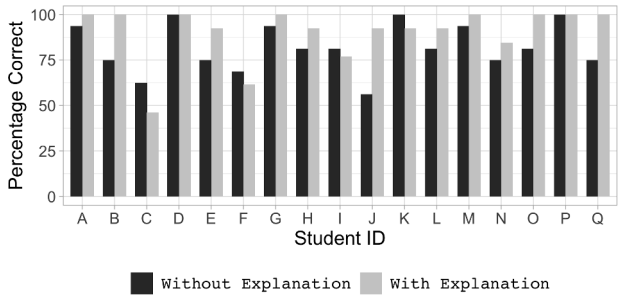

Fig. 3. Graph of the student percentages scores of questions for formula without (black) and with (grey) explanations.

time, and the first student left after 40 minutes.

Eighteen (18) students completed the questionnaire. Of these, we gained informed consent from 17 students; thus, we exclude the data of one student. Informed consent was collected during a previous class by one of the authors without the presence of the course instructor. Students were assured that non-participation would not affect their standing in the course. By separating the consent process from the study itself, we achieved a more natural class setting.

No participant had prior experience with LTL, but one student (P) indicated awareness of other notations from work in the industry. All questions were voluntary (i.e., no question was mandatory). With the exception of Student Q, who started fifteen minutes after the rest of the participants, all students answered all questions. Student Q skipped the questions for formula H (see Table I) and instead completed Period 3.

Data Processing. After the in-class session, the collected data was anonymized and divided into questions with a definitive answer, qualitative response, and student feedback (see codebook online[1]). The qualitative questions were independently adjudicated for correctness by two authors. The authors agreed on 92% of the responses (achieving moderate inter-rater reliability $\kappa = 0.57$ using Cohen's method [28]). Conflicts between the two adjudications were then evaluated by a third author and discussed among all the authors. The adjudicated qualitative data was then aggregated with the questions with definitive answers to create the students score on the educational intervention (i.e., one point per question). The student feedback collected in Period 3 was analyzed by one author and verified by a second author.

B. Results

RQ1: LLM Explanations and Scores. We evaluated students understanding of the formulae in Table I, and students performed well overall with a median score of 29 (IQR 27-32). The highest score was 33/33 (Student P), and the lowest score was 19/33 (Student C). Formula F had the greatest percentage of correct responses (97% correct), while formula H had the least correct responses (72% correct without Student Q).

Null Hypothesis 1: There is no observable difference between the median of the distributions of students' scores with and without the LLM explanations.

We compared the scores with and without an associated explanation. Students' scores were higher on formulae with an explanation. This phenomenon exists with and without the LLM-Belief questions. Fig. 3 shows the percentage of correct responses for the formulae questions (without the LLM-Belief questions) of each student, with and without explanations. Two students (D, P) scored equally well with perfect scores, and three students scored better without the explanations (C, F, K[2]); the remainder all favored better with the explanations. We find this difference in score to be significant at the $\alpha = .05$ level ($p = .032$, Wilcoxon paired signed rank test), rejecting Hyp. 1. The median improvement in students' scores is 6% (IQR 0-17%) with the explanation, meaning most students improved by at least one question, although we do not make claims about the true magnitude of the effect size since we are comparing percentages because the total number of questions was not the same for each formula (see Sect. IV-A). **To answer RQ1, we found a difference between students' scores with and without explanations, indicating an effect on student understanding.**

RQ2: Helpfulness of Explanations. We directly asked students if they found the LLM explanations helpful. Fig. 4 visualizes to what extent students found the explanations helpful in understanding the LTL formulae. The vast majority of students (15/17 or 88%) found the explanations at least moderately helpful. The two students (E and J) who found the explanations only slightly helpful both performed better with the explanations, with Student J having the largest percentage difference between the two measures (see Fig. 3). Further, we asked students how they would improve the explanations, and five (5) students thought the descriptions were good and didn't make suggestions about how to improve them. The twelve remaining students wanted one or more changes: five (5) wanted more examples, five (5) wanted more visualizations, three (3) wanted them to be shorter, and two (2) wanted more detailed breakdowns. **We find that students preferred formulae with explanations, with roughly 40% of them finding the explanations helpful without any recommended modifications, answering RQ2.**

RQ3: Students' Perceptions. In Period 3 of the intervention, we gauged the students' subjective experiences. Three-quarters of students reported learning LTL as similarly challenging to other course topics, with the remainder reporting it as less challenging (no one reported it as more challenging). When asked which temporal operator was the most challenging to understand, eight (8) students found Until U to be the most difficult temporal operator. Of the remaining students, four (4) selected Eventually F, four (4) selected Globally G, and one (1) selected Next X. We also inquired about what aspects made the questions challenging to answer, gave a few examples, and allowed students to write independently. From this data, we find that eleven (11) selected "interpreting the implications of formulae" as the hardest part, while eight (8)

[2]Students C and F had the lowest overall scores. Student K missed only one question on the assessment.

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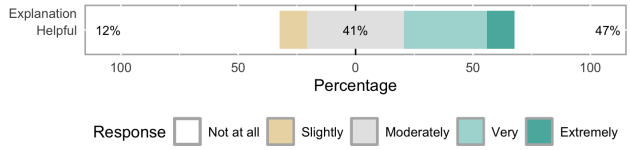

Fig. 4. Graph of student responses to the question, "to what extent were the explanations helpful in understanding the LTL formula?".

selected "applying the formula to sequences", with five (5) students selecting both of these. Only one student (B) selected "none, I did not find the questions challenging".

To understand engagement and whether students saw the applicability of LTL, we asked them how they could "use LTL (or this kind of reasoning) in a computer science project?". Ten (10) students produced answers that showed clear applicability. For example, students reported examples such as video games, autonomous cars, simulations, and software with timed events. Student G explained how each temporal operator would be used. The remaining seven (7) students were either unsure or gave vague answers (e.g., "by simplifying the logic behind some algorithms" [Student L]). **To summarize our data for RQ3, students perceived the LTL intervention as an appropriate level of difficulty for their class and varied in which parts they found most challenging.**

C. Discussion and Limitations

Qualitative Observations. While we invited students to use outside resources, none of the students appeared to use any other resources than the lecture slides and cheatsheet. In Period 3, multiple students noted that they had asked either the student researcher or instructor a question. During Period 2 of the study, students were confused with how to think through true/-false questions about the satisfiability of an example sequence, how to write infinite sequences, and what the meaning of the first value in the sequence is. For example, Student P recommended changing the wording of the question from "[i]n a sequence where..., the formula is violated" to "the formula is *always* violated". Future iterations should add more concrete explanations of the notation around sequences of states.

As mentioned in Sect. III, our prompt was necessarily insufficient. For example, in the explanation for $F(!e)$ (i.e., Formula B) and $G(!a \rightarrow F\ z)$ (Formula G), the LLM explanation identified the breakdown as "F(...): "Finally" (F) — at some point in the future"; yet, F stands for *eventually*. The resulting refined phrasing was correct in both cases, but this may have caused confusion about using F. Two students (E and N) correctly identified this error in terminology.

Identifying Points of Confusion. Students particularly struggled with the Until (U) operator, which eight students identified as the most difficult temporal concept to grasp. Eleven students cited "interpreting the implications of formulae" as their primary challenge, while eight found difficulty "applying the formula to sequences". Students also expressed confusion about when temporal constraints begin and end, particularly in nested expressions. One participant questioned "whether F would be separate for each instance" [Student A] and how a single satisfaction of F impacts multiple trigger conditions. Additionally, students requested improvements to the explanations, with five wanting more examples, five seeking better visualizations, and three preferring more concise descriptions.

Threats to Validity. Our study is only a preliminary investigation and was limited by small sample size and educational constraints, which prohibited a control group and limited interactions with students to one class session. Learning and carry-over effects likely occurred as students progressed through the formulae. Our only attempt to mitigate this was to vary which formulae included explanations across different difficulty levels. Within each pair (e.g., C/D and G/H) and across the sequence, students may have learned about the operators from the explanations. Both of these effects may have confounded our results for RQ1, and further investigation is required to verify our claims, as well as more precisely measure their effect size. Additionally, the constructs of helpfulness and understanding are threatened by our use of a unipolar scale and our tool-agnostic design. Students may perform differently when interacting directly with an LLM, and may have perceived the helpfulness of the explanations differently if presented with multiple levels of unhelpfulness [29]. The single-session format also meant students may have experienced fatigue toward the end of the session, potentially affecting their performance on more complex formulae. Moreover, the voluntary nature of participation could have introduced some self-selection bias, as students more interested in formal notations may have shown more engagement. While students are one of our target populations for this work, CS undergraduates may not generalize to the broader population of stakeholders. The classroom setting and relatively simple LTL formulae may not fully capture the complexity of interpreting formal notations in all industrial contexts, and the helpfulness of our LTL explanations may not scale to complex nested formulae. Our focus on specific LTL temporal operators may not generalize to other formal notations or more complex formula structures. Our small sample size (17 students) limits the statistical power of our analysis. The subjective evaluation of student responses also introduces potential measurement variability, which we attempted to mitigate through independent evaluation by multiple researchers.

V. Updated Prompt for Generation

Next, we describe our methodology for creating our improved prompt for LLM-based explanations of LTL formulae.

Incorporating Points of Confusion. Our initial prompt (see Sect. III) can be improved by incorporating points of confusion. First, we integrate the points we discussed in our classroom study (see Sect. IV-C for details). Second, as part of creating the gold standard for the 50F dataset (see Sect. III), we identified common misconceptions. For example, explanations may not describe when temporal constraints begin and end, the satisfaction conditions for complex operators, the implications for state transitions, and satisfying and violating sequences.

For state transitions involving Until (e.g., $p\ U\ q$), explanations frequently fail to emphasize that once q becomes true,

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the entire formula is satisfied regardless of the value of p. When explaining implication (e.g., $a \rightarrow b$), explanations may state only that "if a holds, then b must hold" without making explicit what happens when a does not hold. Further, when breaking down more complex formulae (e.g., $(d \mid !b)\ U\ !a$), the explanation may list conditions randomly instead of prioritizing the part that determines when the formula terminates.

Prompt Refinement. The updated prompt template is initialized with system contextualization. This involves defining the LLM's role as an expert in formal RE, particularly in Linear Temporal Logic, aiming to draw attention to relevant knowledge that the LLM may already have in this domain.

> *You are an expert in formal requirements engineering with a focus on Linear Temporal Logic (LTL). Your goal is to translate formal LTL formulae into natural language, providing clear, precise explanations that account for when the formula is satisfied and when it imposes constraints on the system over time.*

While our original approach focused on structural decomposition through parse trees, our analysis showed that effective LTL explanations need to address temporal semantics and constraint boundaries more explicitly and provide clearer distinctions between when constraints apply. For example, we instruct the LLM to "distinguish between the immediate satisfaction of the formula and the ongoing constraints that apply over time". Next, we provide specific guidelines for explaining each temporal operator, with particular emphasis on their semantic meanings and constraint boundaries, which address the points of confusion discussed above.

We retained the one-shot prompting paradigm, guiding the LLMs reasoning with an example. The updated example gives a complete reasoning process for the Until (U) operator—which students found most challenging—demonstrating precisely how constraint boundaries should be explained. Specifically, we use the formula $(c\ \&\ b)\ U\ a$ in our prompt, recommending the following translation:

> *The translation would be: a has to be true either in the moment in which the formula is evaluated (and in that case the formula does not constrain the value of c and b) or at some point in the future (and in that case, c and b have both to be true from the moment the formula is evaluated until the instant before a is true). The formula does not constrain the value of c and b in the moment in which a is true or after it.*

This approach leverages the LLM's ability to recognize patterns and apply similar reasoning structures across different formulae while remaining focused on the semantic nuances of temporal logic. We strategically placed this example at the end of the prompt rather than the middle, as LLMs have been shown to exhibit U-shaped attention bias and the order in which samples are provided can make the difference between near state-of-the-art and random guess performance [30]. This cognitive phenomenon causes LLMs to allocate significantly greater attention to the information presented at the beginning and end of a prompt sequence while showing reduced retention and processing capacity for content positioned in the middle sections. By doing so, we ensure that our critical example would receive processing priority. The full text of our updated prompt is available online[1].

VI. Evaluation of Explanation and Summary Quality and Correctness

In this section, we analyze the quality of the explanations and summaries generated by the LLMs and compare them with our gold standard and student responses. We ask three research questions:

RQ4 To what extent has the resulting explanations' quality changed with our updated prompt?

RQ5 How does the quality of the LLM-generated formula summaries compare with summaries written by students and experts?

RQ6 How do professionals rate the quality of the summaries?

Operationalization of Quality Construct. Before answering research questions RQ4 and RQ5, we must first operationalize our construct of explanation quality. We reviewed the linguistics literature and defined quality as three criteria: (1) Fluency & Readability [31]: Does the sentence sound natural and smooth? (2) Coherence & Logical Flow [32]: Does it maintain a clear structure and make sense? (3) Lexical & Syntactic Quality [33]: Is the word choice, grammar, and structure appropriate?

We chose to use the Flesch Reading Ease score (FRE) [34] and the Dale-Chall (DC) score [35] to computationally evaluate Fluency & Readability. After considering the many scoring systems available in the literature, we decided to compare these two because they are examples of the two different themes of metrics. The FRE listed in Def. 3 is directly computed based on the total number of words, syllables, and sentences. The highest FRE score is 121 and negative scores are possible, although most texts fit within the range of 10 to 100. A lower score indicates a higher reading level, with 60–70 considered as around an eighth or ninth-grade reading level. Unlike the FRE score, the Dale-Chall score compares the words present in the document with a list of easily understood words. It then computes a score indicative of grade level, ranging from 4.9 or lower (readable by fourth graders and lower), to 9.9 (college student understanding).

$$206.835 - 1.015\left(\frac{\textit{total words}}{\textit{total sentences}}\right) - 84.6\left(\frac{\textit{total syllables}}{\textit{total words}}\right) \quad (3)$$

To compute Coherence & Logical Flow, we developed a custom metric that measures paragraph coherence based on documentation of the TERA [36], [37] tool, which measures coherence using the Coh-Metrix [38]. Our approach quantifies the logical flow and semantic consistency between sentences by first extracting individual sentences from each paragraph and generating context-aware embeddings using a pre-trained BERT model [39]. These embeddings capture deep semantic relationships beyond simple word overlap. We then compute pairwise cosine similarity scores between sentence embeddings and average them to produce an overall coherence score. This score ranges from -1 to 1, with higher values indicating

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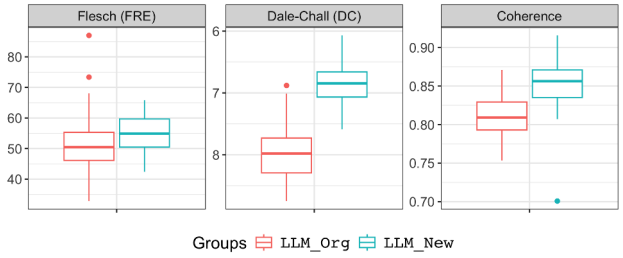

Fig. 5. Box plots of the Flesch Reading Ease (FRE), Dale-Chall (DC), and Coherence scores for the full explanations of each formula in the 50F dataset, using the original (LLM_Org) and updated (LLM_New) prompts.

stronger coherence. This metric can only be computed for the explanations (RQ4) and not the summaries (RQ5) because most summaries contain only one sentence.

We were unable to find an appropriate computational method to evaluate Lexical & Syntactic Quality and argue that it would not make sense in this case because the particular word choice, grammar, and structure must match the LTL formula. Instead, we used this metric to evaluate the correctness of the LTL explanations relative to their formula through expert judgment for RQ4, and the general lexical and syntactic quality of the summaries with professionals for RQ6.

RQ4: Changes in Explanation Quality. With our definition of quality described above, we can now address RQ4, how the quality of the resulting formula explanations changed with our updated prompt (see Sect. III for definitions).

Null Hypothesis 2: There is no observable difference between the quality scores based on whether they were generated with the original or updated prompt.

We computed the Flesch Reading Ease (FRE), Dale-Chall (DC), and the Coherence scores for the 50 formula dataset using the original prompt discussed in Sect. III (identified as LLM_Org) and the updated prompt discussed in Sect. V (identified as LLM_New). Fig. 5 shows box plots for each of the scores, grouped by LLM_Org and LLM_New. Recall the scales of FRE and DC are reversed, meaning a higher score on FRE and a lower score on DC indicate easier fluency and readability. From the box plots in Fig. 5, it is evident that the median score for the updated prompt (LLM_New) is better than the original prompt (LLM_Org) across all three measures, accounting for both Fluency & Readability and Coherence & Logical Flow. We find that the difference in scores is significant at the $\alpha = .05$ level across all three scores (FRE: $p = .012$, DC: $p < .001$, and Coherence: $p < .001$, via the Wilcoxon paired signed rank test), rejecting Hyp. 2. We did not compare the Lexical & Syntactic Quality of the two prompts. Instead, one author evaluated each explanation in the dataset generated by the updated prompt (LLM_New) for correctness. We found 84% (42/50) of the formula explanations to be correct. **We conclude that our updated prompt produces better quality explanations, answering RQ4.**

RQ5: Comparing Machine and Human Summaries. Next we investigate RQ5, where we compare the formula summaries (see Sect. III) generated by the LLM with those written by humans (i.e., students and experts). In this research question, we only consider Fluency & Readability because the summaries are short, and therefore, the notion of Coherence & Logical Flow does not apply. To operationalize Fluency & Readability, we again use the Flesch Reading Ease (FRE) and the Dale-Chall (DC) scores. We compare the summaries given by the original and updated prompts (LLM_Org and LLM_New, respectively) directly with the summaries in our gold standard (Gold) for the 50F dataset. Fig. 6 shows box plots for each of the FRE and DC scores, grouped by data source. In addition to the scores of the 50F dataset, we include the scores for all student responses (Students) in our classroom study (see Sect. IV) to all questions where students were asked to write summaries for a given formula (see question wording in Sect. III). In comparing LLM_Org and LLM_New, we see a similar trend as in RQ4 where the updated prompt (LLM_New) produces more readable results than the original prompt (LLM_Org); yet, here we investigate how the LLM outputs compare with those generated by humans. The results of the Flesch Reading Ease (FRE) scores show variations between groups of humans. For example, the gold standard (Gold) created by an LTL expert had a median FRE score of 63 (IQR 34-68) whereas the students' responses (Students) had a median FRE score of 84 (IQR 74-93), meaning the expert used a higher fluency (or grade-level) than the students. The updated prompt (LLM_New) produced a median of 68 (IQR 62-75), which was between the results of the Gold standard and the Students. However, the results of the Dale-Chall (DC) scores showed similarity between groups. The medians of LLM_New, Gold, and Students were 7.8 (IQR 7.4-8.3), 7.7 (IQR 7.2-8.4), and 7.6 (IQR 7-8.4), respectively. For completeness, the medians of LLM_Org for FRE and DC were 58 (IQR 43-71) and 8.8 (IQR 8.3-9.7), respectively. **Therefore, we found mixed results for RQ5. Depending on the measure, the refined fluency of the updated prompt was similar to humans (i.e., between novice students and an expert).**

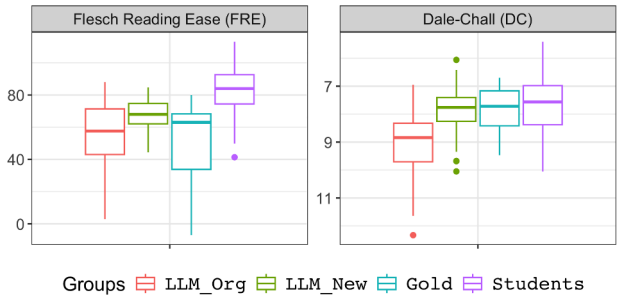

Fig. 6. Box plots of the Flesch Reading Ease (FRE) and Dale-Chall (DC) scores for the summaries of each formula in the 50F dataset, generated by the original (LLM_Org) prompt, the updated (LLM_New) prompt, and included in our gold standard. FRE and DC scores of all student responses from the classroom study discussed in Sect. IV included for benchmarking purposes.

RQ6: Human Judgement. Finally, we conclude our analysis by investigating RQ6: How do professionals rate the quality of the summaries? We recruited volunteers (i.e., *Readers*) within our professional network to qualitatively evaluate the

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summaries. Although the volunteers were not meant to be representative of all stakeholders involved in any project, each reader represented possible areas of expertise and domain knowledge within collaborative projects: (1) requirements engineering (paper author), (2) psychology, (3) electrical and computer engineering, (4) education and psychology, and (5) business and anthropology. Only Reader 1 had any prior exposure with LTL—the remaining readers were evaluating the summaries based solely on their English understanding. Each reader was asked to evaluate each of the three quality criteria using a unipolar scale: (1) Not at all, (2) Slightly, (3) Moderately, (4) Very, and (5) Extremely/Completely. Readers 1 and 2 evaluated the entire dataset, whereas Readers 3–5 evaluated roughly half the dataset. To mitigate the threat of *experimenter expectancies* and *hypothesis guessing*, one author used a script to randomize the data (see online[1]) and then anonymized the data. After each reader completed their assessment, the same author de-anonymized and sorted the data for analysis.

Table II lists the aggregate analysis for each data source and reader. For example, the first row gives the counts for each point on our five-point Likert-type scale for Reader 1 on the LLM_Org data. No formula received a score of 1 for Fluency & Readability or Coherence & Logical Flow. From Table II, we see that the readers' scores were all quite high. The median score across all three quality criteria was 5 (IQR 4-5). Looking at trends in the data in Table II, the gold standard and the student summaries were rated to be of lower quality than either of the LLM-generated summaries, with the student summaries rated worse for Fluency & Readability. LLM_Org had the highest rating across the three criteria, outperforming LLM_New. The two readers with training in engineering (i.e., J1 and J3) tended to be more critical than the other readers (see below for further discussion). **Thus, professionals rated the quality of the summaries in this study to be high, with LLM generated summaries rating more favorably than human-generated summaries, answering RQ6.**

Discussion, Limitations, and Threats to Validity. In RQ4, we found that updating our prompt improved the Fluency & Readability and Coherence & Logical Flow based on the FRE, DC, and Coherence scores. Based on the metrics used in RQ5, the updated prompt was either at the level of fluency of novices or in between experts and novices. However, in RQ6, we found that professionals rated the original prompt the best and the LLM-generated summaries more favorable. This conflicting evidence warrants further examination and research. We found the empirical evidence of RQ4 and RQ5 to be compelling and think the updated prompt produces better explanations and summaries. Our new hypothesis is that the original prompt produced more generic and lower-quality sentences from the perspectives of understanding LTL, but these same sentences appeared easy to read when read by non-experts. This hypothesis stems from our observation that the readers with CS training were more harsh, and we theorize that they were aiming for a deeper understanding of the content than those readers without a background in CS. Since the goal of our research is to help non-experts and students improve their understanding of LTL formulae, we argue that the shift in simplifying the resulting explanations generated by our updated prompt is an improvement and instead we chose the wrong target population for our investigation in RQ6, which threaten the validity of these results. By using our professional network to recruit readers, we biased our sample towards individuals with advanced degrees and, by extension, higher levels of comprehension. Finally, further investigations of quality should use the entire explanation rather than the summaries and investigate individuals' ability to interpret the formula rather than solely review the summary text.

Our evaluation of explanation quality relied on readability metrics (e.g., FRE), but these may not fully capture all aspects of effectiveness. While these automated measures assess sentence structure and complexity, they do not account for semantic accuracy or how well an explanation supports learning. When comparing the original and revised prompts, other factors could have influenced our results, such as variations in LLM performance over time.

VII. Discussion and Limitations

In Sect. IV-C, we discussed the specific limitations of our classroom study. In Sect. VI, we gave a brief overview of the limitations of our quality study. In this section, we take a step back and look at the promise, limitations, and future directions of our work more generally.

The classroom evaluation study, while providing valuable insights into the effectiveness of LLM explanations, also revealed several pedagogical limitations concerning the foundational understanding of LTL and the nature of the examined formulae. As noted in Sect. IV, the examined LTL formulae were relatively simple. Consequently, while the prompt was designed to elicit explanations that break down LTL formulae into comprehensible parts and emphasize the reasoning process, for these simpler formulae, the explanations might have appeared to students as providing direct solutions rather than extensively guiding them toward independently understanding and translating more complex formulae. In future interventions, the provided LTL materials could be more comprehensive, for example, clarifying operator precedence and associativity or introducing common equivalences.

In Sect. III, we introduced the 50F dataset and our gold standard. As part of our analysis for RQ6, Reader 1 (an LTL expert) rated three summaries from the gold standard lower on the Lexical & Syntactic Quality scale (see scores 1/2 in Table II). Upon investigation, we found that 3/50 summaries in our gold standard contained errors, which were likely the result of fatigue and the repetitive nature of the original creation process. These errors highlight the importance of verifying our gold standard, and suggest that even experts are not infallible and may benefit from external support. Additionally, while our dataset was diverse, it may not fully represent the complexity of real-world specifications. Future work can explore including domain-specific information as part of the formulae

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TABLE II
COUNTS OF READERS SCORES

Source	Reader	Total	Fluency & Readability 1	2	3	4	5	Coherence & Logical Flow 1	2	3	4	5	Lexical & Syntactic Quality 1	2	3	4	5
LLM_Org	J1	50	0	0	3	20	27	0	0	2	8	40	0	3	11	16	20
	J2	50	0	0	3	4	43	0	0	2	5	43	0	0	2	5	43
	J3	24	0	0	2	5	17	0	0	3	3	18	0	0	1	1	22
	J4	24	0	0	0	0	24	0	0	0	0	24	0	0	0	0	24
	J5	25	0	0	1	10	14	0	0	1	8	16	0	0	4	6	15
LLM_New	J1	50	0	0	14	7	29	0	2	2	12	34	2	4	5	14	25
	J2	50	0	0	3	5	42	0	0	3	5	42	0	0	0	11	39
	J3	25	0	1	3	6	15	0	1	4	5	15	0	2	6	6	11
	J4	25	0	0	0	0	25	0	0	0	0	25	0	0	0	0	25
	J5	25	0	0	4	7	14	0	0	2	8	15	0	0	3	6	16
Gold Standard	J1	50	0	0	1	3	46	0	0	1	3	46	1	2	1	2	44
	J2	50	0	0	7	8	35	0	0	7	8	35	0	0	2	13	35
	J3	25	0	5	4	7	9	0	1	6	6	12	0	0	7	6	12
	J4	25	0	1	3	4	17	0	0	3	6	16	0	0	1	7	17
	J5	25	0	0	5	12	8	0	0	6	6	13	0	1	4	9	11
Students	J1	21	0	1	3	12	5	0	0	2	6	13	2	1	10	3	5
	J2	21	0	0	0	2	19	0	0	1	1	19	0	0	0	3	18
	J3	10	0	0	1	3	6	0	0	0	4	6	0	0	1	3	6
	J4	11	0	0	0	1	10	0	0	0	1	10	0	0	0	1	10
	J5	11	0	0	3	5	3	0	0	2	5	4	0	1	2	2	6

For each Reader (J1-J5), counts represent the total number of summaries evaluated at a specified level (1–5) within each category (e.g., Fluency & Readability). Separate tables are given for each data source (e.g., Students). The Total column represents the number of summaries that each reader evaluated for each source. The levels are (1) Not at all, (2) Slightly, (3) Moderately, (4) Very, and (5) Extremely/Completely.

and prompt. For example, our investigation only used single letters as propositions rather than contextual information, such as Def. 1 and Def. 2 in our microwave example in Sect. I.

Humans and LLMs have trouble translating formulae when they are not written in a natural way (e.g., $!X\ !(!g \mid !j)$); yet, these formulae can happen in the real world, especially when automatically generated by tools. In this case, future work could explore integrating our work with deductive methods to simplify the formulae and reason on their truth value. Further, some explanations generated by the LLM prompts were long and repetitive. These highly detailed explanations would be of use to novices and those learning LTL, but would be undesirable in an industry setting. As future work, we imagine personalized prompts for different purposes.

In this paper, we applied our approach to LTL, but our ultimate goal is to create a framework that can be used to generate explanations for a variety of formal languages and artifacts. In our initial approach, we started with easier formulae and elicited user feedback early in our process (via our classroom study, see Sect. IV). We iterated over our prompt and integrated both expert and novice feedback at multiple stages and with multiple groups (including non-technical professionals, see Sect. VI), and this approach proved successful for LTL. In future work, we will explore how to use these steps to create a generalizable framework for formal notations. We can incorporate our framework into the vision presented by Ferrari and Spoletini [8], where our LLM-based explanations are just one part of their full cycle of FM-based development, including verification.

VIII. RELATED WORK

LLMs for RE Tasks. LLMs have been used for elicitation, classification, and specification generation in RE. Ronanki et al. assessed ChatGPT's ability to generate software requirements, finding that it produces highly abstract and consistent requirements that align with structured RE processes [40]. Hymel and Johnson [41] demonstrated that LLMs generate aligned and complete requirements faster and at a lower cost than humans; but, found that the models struggle with domain-specific nuances. Other work has shown that LLMs can identify dependencies and inconsistencies in large-scale requirement datasets with reasonable accuracy [42], [43]. However, challenges remain in distinguishing between requirements relationships and traceability across evolving specifications [44].

LLMs and Formal Languages. Recent research has explored the capabilities of LLMs in working with formal languages. For example, Chen et al. [9] and Cosler et al. [10] show that LLMs can effectively generate formal logic representations with minimal training; yet, challenges remain in ensuring syntactic validity and logical coherence, as LLMs may generate incorrect or unverifiable specifications. To address these limitations, hybrid approaches combining pipeline decomposition and majority-voting mechanisms have been proposed [45], demonstrating accuracy improvements over both rule-based and end-to-end neural models. While LLMs approach human-level comprehension, they still struggle with accurately generating logical forms [46]. To enhance LLM reasoning for complex tasks, various prompting strategies have been developed that leverage intermediate steps. For instance,

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CoT prompting facilitates reasoning by eliciting a series of intermediate steps, and, building on this, Plan-and-Solve (P+S) prompting [20] extends zero-shot CoT by explicitly instructing the LLM to devise its own plan for problem decomposition, aiming to address issues such as missing-step error. Furthermore, a more complex and computationally intensive paradigm involves multi-agent debate (MAD) [19], where multiple LLM instances collaboratively reason and reach consensus through a debate process. In contrast, our approach employs a more prescriptive, structured prompt. Rather than relying on the model to devise its own reasoning structure or engage in a multi-agent debate, our method explicitly provides a predefined format and sequence for these intermediate steps within the prompt itself. Our guided decomposition strategy ensures a methodical and comprehensive breakdown of complex LTL formulae, which is crucial for achieving precise and logically coherent explanations in the domain of formal languages. Our refined prompt defines the LLM's role as an expert in LTL, aiming to draw attention to relevant knowledge the LLM may already possess in this domain.

Ferrari et al. [8] outlined how LLMs can enhance RE activities, emphasizing their role in bridging between natural and formal languages. While ensuring correctness remains a challenge, recent work suggests that hybrid approaches, which combine LLMs with formal verification techniques, could mitigate these limitations [47]. Saleem et al. note that LLMs under perform in structured extraction and classification tasks when compared to domain-specific approaches [48], while Sami et al. show how prompt engineering can play a crucial role in improving LLM performance within a structured RE workflow [49]. Our research builds upon this foundation, specifically examining how LLMs can be used to make formal languages more accessible to non-experts.

LLM-Generated Explanations for Software and RE Artifacts. LLMs can provide automated explanations for requirements specifications, aiding in traceability and validation tasks. Fuggitti and Chakraborti incorporate verification into their LLM translation of natural language descriptions into LTL formulas [50]. Several studies evaluate LLM-generated explanations for structured requirement formats. With well defined templates, LLMs can support validation tasks [51], and provide contextualized justifications for changes in evolving requirement documents [52].

LLMs have also been explored in software development. For example, Ali et al. [53] propose combining retrieval-augmented generation with LLMs to establish traceability links between NL requirements and software artifacts. Research in LLM-based test case analysis examines structured explanation generation for software testing, particularly in security-critical software [54]. Meanwhile, CodeEditor [55] offers insights into code transformations, dependency resolutions, and bug fixes by integrating LLMs within IDEs.

Our research builds on these efforts by investigating how LLMs can enhance the explainability of formal specifications, with a focus on correctness and stakeholder comprehension.

IX. Conclusions and Future Work

In this paper, we developed an initial and an improved prompt to generate explanations of LTL formulae. We evaluated using generated explanations as a learning aid in the classroom, and found that students performed better when provided with an explanation and preferred having explanations while learning LTL. Using the gold standard dataset we developed, we compared summaries generated with our initial and improved prompt against those written by students and our gold standard. Our analysis showed that the generated summaries were linguistically appropriate for the target population. Therefore, we conclude that using LLMs to automatically generate explanations of LTL formula is a promising direction of research and worthy of future exploration.

Our work makes several novel contributions in the intersection of formal methods and AI-assisted RE. While prior work has explored LLM capabilities in writing formal specifications, this study is among the first to evaluate the use of LLM-generated explanations of formal notations. The potential impact of this research extends beyond educational settings to industrial practice, where formal methods remain underused, despite their benefits. By demonstrating that LLMs can help bridge the communication gap between formal specifications and non-expert stakeholders, our work could increase adoption of formal methods in multi-disciplinary teams, enhance requirements validation processes, and ultimately improve software quality without sacrificing the precision that makes formal methods valuable.

In future work we continue to improve our prompt and explore which language models are best for the task of explaining formal notations. For example, we consider adding multiple demonstrations and example sequences to our prompt, as well as how to alter the formatting of the resulting explanation. To further validate our approach, we investigate the effectiveness of the updated prompt. In one such study, we repeat the classroom study described in Sect. IV with the updated prompt in isolation. In another study, we examine the scalability of the prompt on nested formulae. Additionally, we explore how direct interaction with the LLM alters the experience of our target population, as well as how hallucinations or incorrect LLM explanations affect learning. We will extend our work beyond LTL to other notations and broaden our study population. For example, we intend to create customized prompts based on stakeholder need and incorporate our work with other formal approaches (see Sect. VII for details). Finally, we continue to investigate improved metrics for Coherence & Logical Flow and Lexical & Syntactic Quality (see Sect. VI).

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REFERENCES

[1] J. M. Wing, "A Specifier's Introduction to Formal Methods," *IEEE Computer*, vol. 23, no. 9, pp. 8–24, 1990.

[2] E. M. Clarke, O. Grumberg, and D. Peled, *Model Checking*. MIT Press, 1999.

[3] A. Hall, "Seven Myths of Formal Methods," *IEEE Software*, vol. 7, no. 5, pp. 11–19, 1990.

[4] J. Woodcock, P. G. Larsen, J. Bicarregui, and J. Fitzgerald, "Formal Methods: Practice and Experience," *ACM Computing Surveys*, vol. 41, no. 4, pp. 1–36, 2009.

[5] B. Greenman, S. Saarinen, T. Nelson, and S. Krishnamurthi, "Little Tricky Logic: Misconceptions in the Understanding of LTL," *The Art, Science, and Engineering of Programming*, vol. 7, no. 2, 2023.

[6] A. Kovetz, N. E. Fuchs, R. Schwitter, and K. Kaljurand, "Controlled Natural Language for Requirements Specification," *Formal Aspects of Computing*, vol. 29, no. 1, pp. 133–162, 2017.

[7] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, Y. Du, C. Yang, Y. Chen, Z. Chen, J. Jiang, R. Ren, Y. Li, X. Tang, Z. Liu, P. Liu, J.-Y. Nie, and J.-R. Wen, "A Survey of Large Language Models," 2023. [Online]. Available: https://arxiv.org/abs/2303.18223

[8] A. Ferrari and P. Spoletini, "Formal Requirements Engineering and Large Language Models: A Two-way Roadmap," *Information and Software Technology*, vol. 181, p. 107697, 2025.

[9] Y. Chen, R. Gandhi, Y. Zhang, and C. Fan, "NL2TL: Transforming Natural Languages to Temporal Logics using Large Language Models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing EMNLP'23*, 2023, pp. 15 880–15 903.

[10] M. Cosler, C. Hahn, D. Mendoza, F. Schmitt, and C. Trippel, "nl2spec: Interactively Translating Unstructured Natural Language to Temporal Logics with Large Language Models," in *Proceedings of the 35th International Conference on Computer Aided Verification (CAV'23)*, 2023, pp. 383–396.

[11] J. Liu, S. Cao, J. Shi, T. Zhang, L. Nie, L. Hu, L. Hou, and J. Li, "How Proficient Are Large Language Models in Formal Languages? An In-Depth Insight for Knowledge Base Question Answering," in *Findings of the Association for Computational Linguistics (ACL'24)*, 2024, pp. 792–815.

[12] K. Thompson, N. Saavedra, P. Carrott, K. Fisher, A. Sanchez-Stern, Y. Brun, J. F. Ferreira, S. Lerner, and E. First, "Rango: Adaptive Retrieval-Augmented Proving for Automated Software Verification," in *Proceedings of the 47th International Conference on Software Engineering (ICSE'25)*, 2025.

[13] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate Before Use: Improving Few-shot Performance of Language Models," in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139, 2021, pp. 12 697–12 706.

[14] P. Spoletini and A. Ferrari, "The Return of Formal Requirements Engineering in theEra of Large Language Models," in *Proceedings of the 30th International Working Conference on Requirements Engineering: Foundation for Software Quality (REFSQ'24)*, 2024, pp. 344–353.

[15] A. Ferrari, F. Mazzanti, D. Basile, and M. H. ter Beek, "Systematic Evaluation and Usability Analysis of Formal Methods Tools for Railway Signaling System Design," *IEEE Transactions on Software Engineering*, vol. 48, no. 11, pp. 4675–4691, 2022.

[16] A. Pnueli, "The Temporal Logic of Programs," in *Proceedings of the 18th IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, 1977, pp. 46–57.

[17] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing," *ACM Computing Surveys*, vol. 55, no. 9, 2023.

[18] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought Prompting Elicits Reasoning in Large Language Models," in *Proceedings of the 36th International Conference on Neural Information Processing Systems (NIPS'22)*, 2022.

[19] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving Factuality and Reasoning in Language Models Through Multiagent Debate," in *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, 2023.

[20] L. Wang, W. Xu, Y. Lan, Z. Hu, Y. Lan, R. K.-W. Lee, and E.-P. Lim, "Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL'23)*, 2023.

[21] K. Valmeekam, A. Olmo, S. Sreedharan, and S. Kambhampati, "Large Language Models Still Can't Plan (A Benchmark for LLMs on Planning and Reasoning about Change)," in *Proceedings of the Foundation Models for Decision Making Workshop*, 2022.

[22] Baier, Christel and Katoen, Joost-Pieter, *Principles of Model Checking*. MIT Press, 2008.

[23] H. Cherukuri, A. Ferrari, and P. Spoletini, "Towards Explainable Formal Methods: From LTL to Natural Language with Neural Machine Translation," in *Proceedings of the 28th International Working Conference on Requirements Engineering: Foundation for Software Quality (REFSQ'22)*, 2022, pp. 79–86.

[24] A. van Lamsweerde, *Requirements Engineering - From System Goals to UML Models to Software Specifications*. Wiley, 2009.

[25] R. Wieringa, *Design Science Methodology for Information Systems and Software Engineering*. Springer, 2014.

[26] S. Min, X. Lyu, A. Holtzman, M. Artetxe, M. Lewis, H. Hajishirzi, and L. Zettlemoyer, "Rethinking the Role of Demonstrations: What Makes In-Context Learning Work?" in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 11 048–11 064.

[27] OpenAI, "GPT-4o Mini: Advancing Cost-Efficient Intelligence," 2025, available online at *https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/*, accessed 03/08/2025.

[28] J. Cohen, "A Coefficient of Agreement for Nominal Scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.

[29] P. M. Bentler, "Semantic Space is (Approximately) Bipolar," *Journal of Psychology*, vol. 71, no. 1, pp. 33–40, 1969.

[30] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the Middle: How Language Models use Long Contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, 2024.

[31] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A Method for Automatic Evaluation of Machine Translation," in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, 2002, pp. 311–318.

[32] R. Barzilay, N. Elhadad, and K. R. McKeown, "Inferring Strategies for Sentence Ordering in Multidocument News Summarization," in *Journal of Artificial Intelligence Research*, vol. 17, 2002, pp. 35–55.

[33] S. Banerjee and A. Lavie, "METEOR: An Automatic Metric for MT Evaluation with Improved Correlation with Human Judgments," in *Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization*, 2005, pp. 65–72.

[34] R. Flesch, "A New Readability Yardstick," *Journal of Applied Psychology*, vol. 32, pp. 221–233, 1948.

[35] E. Dale and J. S. Chall, "A Formula for Predicting Readability," *Educational Research Bulletin*, vol. 27, no. 1, pp. 11–28, 1948.

[36] G. T. Jackson, L. K. Allen, and D. S. McNamara, *Common Core TERA Common Core TERA: Text Ease and Readability Assessor*. In "Adaptive Educational Technologies for Literacy Instruction", Routledge, 2016, p. 49.

[37] "The Coh-Metrix Common Core Text Ease and Readability Assessor (T.E.R.A.)," Available online at *https://soletlab.adaptiveliteracy.com*, accessed 02/15/2025.

[38] D. S. McNamara, A. C. Graesser, P. M. McCarthy, and Z. Cai, *Automated Evaluation of Text and Discourse with Coh-Metrix*. Cambridge University Press, 2014.

[39] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*, Jun. 2019, pp. 4171–4186.

[40] S. Ronanki and R. Gupta, "Can Large Language Models Generate Requirements? An Empirical Evaluation," 2023. [Online]. Available: https://arxiv.org/abs/2307.07381

[41] S. Hymel and T. Johnson, "LLMs vs. Human Experts: A Comparative Study in Requirements Elicitation," in *Proceedings of the 46th International Conference on Software Engineering (ICSE'24)*, 2024.

[42] S. Abualhaija *et al.*, "Automating Requirements Traceability with Large Language Models: Challenges and Opportunities," *Journal of Requirements Engineering*, 2023.

Authorized licensed use limited to: University of Illinois at Chicago Library. Downloaded on February 09,2026 at 05:01:09 UTC from IEEE Xplore. Restrictions apply.

[43] I. Vlasov *et al.*, "Investigating LLMs for Automated Requirements Classification and Refinement," *IEEE Transactions on Software Engineering*, 2023.

[44] S. Winkler *et al.*, "Large Language Models for Requirements Engineering: A Systematic Assessment," in *Proceedings of the IEEE International Requirements Engineering Conference (RE)*, 2023.

[45] A. Fuggitti *et al.*, "Majority-voting for LTL synthesis: Leveraging Multiple Translation Strategies for Improved Formalization," *Formal Methods in System Design*, 2023.

[46] I. Mirzadeh, K. Alizadeh, H. Shahrokhi, O. Tuzel, S. Bengio, and M. Farajtabar, "Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models," 2024. [Online]. Available: https://arxiv.org/abs/2410.05229

[47] R. Arora and A. Singh, "Prompt Engineering for LLMs in Requirements Engineering: A Systematic Analysis," in *Proceedings of the 30th IEEE International Requirements Engineering Conference (RE)*, 2023. [Online]. Available: https://arxiv.org/abs/2310.13976

[48] M. Saleem, Z. Arshad, and U. Naeem, "Assessing the Role of LLMs in Software Requirements Engineering: Strengths and Limitations," *arXiv preprint arXiv:2412.00959*, 2024. [Online]. Available: https://arxiv.org/abs/2412.00959

[49] R. Sami and D. Lee, "AI-Powered Prioritization in Agile Requirements Engineering," *arXiv preprint arXiv:2405.01564*, 2024. [Online]. Available: https://arxiv.org/abs/2405.01564

[50] F. Fuggitti and T. Chakraborti, "NL2LTL - A Python Package for Converting Natural Language (NL) Instructions to Linear Temporal Logic (LTL) Formulas," in *AAAI Conference on Artificial Intelligence*, vol. 37, 2023, p. 27068.

[51] M. Krishna, B. Gaur, A. Verma, and P. Jalote, "Using LLMs in Software Requirements Specifications: An Empirical Evaluation," *Proc. of 2024 IEEE 32nd International Requirements Engineering Conference (RE)*, pp. 475–483, 2024.

[52] C. Arora, J. Grundy, and M. Abdelrazek, "Advancing Requirements Engineering Through Generative AI: Assessing the Role of LLMs," *Springer, Generative AI for Effective Software Development*, pp. 129–148, 2024.

[53] S. J. Ali, V. Naganathan, and D. Bork, "Establishing Traceability Between Natural Language Requirements and Software Artifacts by Combining RAG and LLMs," *Conceptual Modeling, Lecture Notes in Computer Science*, vol. 15238, pp. 295–314, 2025.

[54] D. Jin, Z. Jin, X. Chen, and C. Wang, "Mare: Multi-Agents Collaboration Framework for Requirements Engineering," 2024.

[55] J. Li *et al.*, "CodeEditor: Learning to Edit Source Code with Pre-Trained Models," *ACM Transactions on Software Engineering and Methodology*, vol. 32, no. 6, September 2023.

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